Wathiq-SPU at AraSeg Shared Task 2026: Calibrated Transformer Ensembles for Arabic Sentence Segmentation across Punctuation and Paragraph Conditions

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Abstract

Arabic sentence segmentation becomes substantially more difficult when punctuation and paragraph boundaries are unavailable. The AraSeg Shared Task 2026 provides a controlled benchmark for studying this problem under four subtasks that progressively remove structural information. We present the Wathiq-SPU submission to all four subtasks of the Closed Track: Paragraph-Aware (PA), No-Punctuation Paragraph-Aware (NoPnx-PA), No-Paragraph (NP), and No-Punctuation No-Paragraph (NoPnx-NP). We formulate sentence segmentation as token-level binary boundary classification using pretrained Arabic Transformer encoders and combine calibrated predictions with subtask-specific ensemble strategies that adapt to the amount of available structural information. CAMeLBERT-MSA serves as our primary encoder, while the most challenging NoPnx-NP subtask additionally benefits from AraELECTRA-based backbone diversity. Our experiments reveal that the most effective source of model diversity depends on task difficulty: diversity among independently trained models provides robust gains when some structural information is available, whereas backbone diversity becomes increasingly important when both punctuation and paragraph cues are removed. The selected systems achieve F1 scores of 93.4, 87.9, 90.0, and 84.8 on the PA, NoPnx-PA, NP, and NoPnx-NP subtasks, respectively.

1 Introduction

Sentence boundary detection is a fundamental preprocessing step in natural language processing and can affect downstream tasks such as translation and information extraction (Gillick, 2009; Read et al., 2012). Although traditional sentence segmentation often relies strongly on punctuation, punctuation alone is not always sufficient, and the problem becomes substantially harder when such cues are missing (Minixhofer et al., 2023; Frohmann et al., 2024).

The AraSeg Shared Task 2026 systematically evaluates this problem for Arabic under different combinations of punctuation and paragraph information (Elkholy et al., 2026b). We participate in all four subtasks of the Closed Track: Paragraph-Aware (PA), No-Punctuation Paragraph-Aware (NoPnx-PA), No-Paragraph (NP), and No-Punctuation No-Paragraph (NoPnx-NP). These subtasks range from the relatively information-rich PA setting to NoPnx-NP, where neither punctuation nor paragraph boundaries are available.

We model segmentation as token-level binary classification using pretrained Transformer encoders, following the broader success of BERT-style contextual representations (Devlin et al., 2019). Our development strategy emphasizes generalization rather than increasingly complex lexical post-processing. We use probability calibration, paragraph structure when available, ensembles of independently trained models, and diversity between pretrained Arabic encoders.

A central finding is that the most useful source of model diversity depends on the difficulty of the subtask. Diversity among independently trained CAMeLBERT models is effective for NoPnx-PA and NP, whereas the hardest NoPnx-NP subtask benefits more clearly from combining CAMeLBERT-MSA with AraELECTRA. We additionally explored meta-classification and feature-based variants, but did not retain them when their improvements failed to remain stable across held-out documents.

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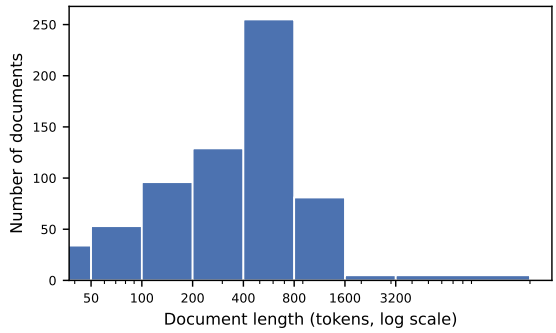


Figure 1: Document-length distribution across the released Train, Dev, and Test splits.

Subtask	Punctuation	Paragraphs
PA	Yes	Yes
NoPnx-PA	No	Yes
NP	Yes	No
NoPnx-NP	No	No

Table 1: Structural information available in each subtask.

2 Background

2.1 Shared Task Description

AraSeg represents each document as an ordered token sequence and requires a binary prediction for every token, indicating whether a sentence boundary occurs after that token (Elkholy et al., 2026b). The released data contain 174 training documents, 222 development documents, and 262 documents in the released test split. A separate Blind Test set is used for the official shared-task evaluation.

The corpus spans eight genres — children's literature, education, literature, media, poetry, politics, religion, and Wikipedia — drawn primarily from the CAMeL Treebank and the Balanced Arabic Readability Evaluation Corpus, with additional religious material from the Quran and Bible translations (Elkholy et al., 2026a). While the majority of the corpus is Modern Standard Arabic prose, the religion and poetry genres include pre-modern and classical registers with markedly different punctuation conventions, further increasing the diversity of sentence-boundary realizations. Document length also varies substantially across the released splits: documents range from 10 to over 18,000 tokens, with a median of 429 tokens and roughly 40% of documents falling between 200 and 800 tokens (Figure 1).

The four subtasks differ in available structural information. PA preserves punctuation and paragraph information; NoPnx-PA removes punctuation while retaining paragraphs; NP preserves punctuation but removes paragraph information; and NoPnx-NP removes both.

2.2 Related Work

Sentence boundary detection has been approached using statistical, rule-based, sequence-labeling, and neural methods (Gillick, 2009; Read et al., 2012; Du et al., 2019). More recent work explicitly addresses segmentation when punctuation is unreliable or absent, though with mixed success outside the languages and domains for which these methods were originally designed. Minixhofer et al. (2023) proposed multilingual punctuation-agnostic sentence segmentation using contextual information, and Frohmann et al. (2024) extended this direction with Segment Any Text (SaT), targeting robustness to missing punctuation and domain variation across many languages. However, when SaT is evaluated directly on AraSeg, it reaches only 53.0 F1 in the hardest, unpunctuated NoPnx-NP subtask, compared with 85.9 F1 for a supervised Arabic-specific encoder (Elkholy et al., 2026b), suggesting that punctuation-agnostic multilingual pretraining alone does not close the performance gap in this Arabic segmentation setting.

Unpunctuated Arabic text has also been studied directly. Alshanqiti et al. (2022) combine a multilingual Transformer with generic linguistic rules to detect punctuation in unpunctuated Arabic, reaching approximately 75% F-measure. Their evaluation, however, targets transcribed, speech-like text segmented into short independent clauses rather than the genre-diverse, sentence-level setting considered here. Their rule-based component was therefore evaluated under a different textual setting, leaving its generalization to classical and religious Arabic unclear. This difference further motivates learning Arabic sentence-boundary evidence directly from genre-diverse supervised data rather than relying on punctuation-restoration rules alone.

Our models build on Arabic-specific pretrained language models, which differ in their training data, morphological preprocessing, and pretraining objective. AraBERT (Antoun et al., 2020) was among the first dedicated Arabic BERT variants, pretrained on large-scale MSA news and Wikipedia text with Farasa-based morphological segmenta-

tion. CAMeLBERT (Inoue et al., 2021) instead offers a family of variant-matched encoders pretrained separately on MSA, dialectal, and classical Arabic, allowing the pretraining register to be matched to the target text; its MSA variant is particularly relevant given AraSeg's predominantly Fus'ha corpus. ARBERT and MARBERT (Abdul-Mageed et al., 2021) follow a similar architecture but diverge sharply in domain: ARBERT targets MSA, while MARBERT is trained primarily on large-scale dialectal social-media text, making it comparatively less matched to AraSeg's formal, genre-diverse register. Unlike these masked-language-modeling encoders, AraELECTRA (Antoun et al., 2021) uses a replaced-token-detection objective, in which a discriminator learns to identify tokens substituted by a generator rather than reconstruct masked tokens. This distinct pretraining objective provides a substantially different representation source from MLM-based CAMeLBERT, motivating its use as a candidate for backbone diversity.

3 System Overview

3.1 Boundary Classification and Ensembling

Each model assigns a sentence-boundary probability p_i to token x_i. The final label is positive when the probability exceeds a subtask-specific decision threshold. For an ensemble of M models, we combine probabilities before thresholding:

$$p_i^{ens} = \sum_{m=1}^{M} w_m p_i^{(m)}, \qquad \sum_{m=1}^{M} w_m = 1. \quad (1)$$

This design allows model training, ensemble weighting, and decision calibration to be optimized separately. Our implementation uses PyTorch and the Hugging Face Transformers ecosystem (Wolf et al., 2020).

3.2 Paragraph-Aware

For Paragraph-Aware (PA), the principal CAMeLBERT predictor is initialized from the strongest No-Punctuation Paragraph-Aware checkpoint and fine-tuned for five epochs with a learning rate of 2×10^{-5}. We use a maximum sequence length of 512 and a stride of 64. The selected system combines the principal predictor with a small auxiliary contribution using weights 0.95 and 0.05. Boundary decisions use a default threshold of 0.543 and a lower threshold of 0.310 for likely sentence-ending punctuation. Paragraph-boundary constraints are applied, and the final document boundary is forced.

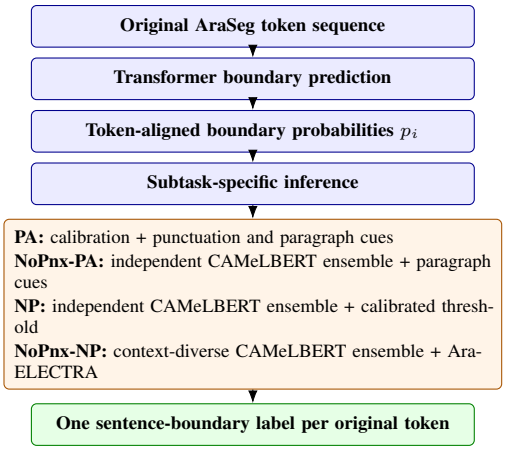


Figure 2: Wathiq-SPU AraSeg pipeline with subtask-specific inference strategies.

3.3 No-Punctuation Paragraph-Aware

For No-Punctuation Paragraph-Aware (NoPnx-PA), two independently trained CAMeLBERT predictors are combined at the probability level using weights 0.40 and 0.60. The selected decision threshold is 0.40. Paragraph structure is represented with `[PAR]`; the marker itself receives label 0, while the immediately preceding token is forced to be a sentence boundary. This allows deterministic paragraph structure to complement punctuation-free contextual predictions.

3.4 No-Paragraph

The final NP system combines three independently trained CAMeLBERT predictors: a principal model and two additional models trained with the same configuration but different random initializations, referred to as Auxiliary Model A and Auxiliary Model B.

Their probability weights are 0.40, 0.30, and 0.30, respectively. The models use a maximum sequence length of 128 and a stride of 16, and the ensemble uses a decision threshold of 0.415. No final sentence boundary is forced.

This configuration was selected through document-level held-out comparison. The global three-model ensemble achieved 0.91833 F1, compared with 0.91686 for the earlier document-adaptive configuration, and performed better in four of five held-out folds. We therefore selected the simpler global ensemble rather than adding decision-model complexity for a small aggregate gain.

3.5 No-Punctuation No-Paragraph

NoPnx-NP was the most challenging subtask. Our strongest CAMeLBERT-only system is a context-diverse ensemble combining a directly fine-tuned NoPnx-NP model (weight 0.10), an NP-to-NoPnx-NP transfer model (0.50), and an auxiliary CAMeLBERT model using a 256-token context (0.40). The combined probability uses a calibrated threshold of 0.445.

To introduce backbone diversity, we fine-tuned AraELECTRA (Antoun et al., 2021). It uses maximum length 256, stride 32, and four training epochs; the third checkpoint was selected using development evidence. The final system combines the context-diverse CAMeLBERT ensemble with AraELECTRA using weights 0.40 and 0.60 and applies a decision threshold of 0.405. No final sentence boundary is forced. We also evaluated other Arabic pretrained encoders and more complex post-processing strategies, including variants based on AraBERT and MARBERT (Antoun et al., 2020; Abdul-Mageed et al., 2021). They did not provide sufficiently stable held-out gains to replace the final CAMeLBERT–AraELECTRA combination.

4 Experimental Setup

We train only on the official AraSeg training split and use the development split for checkpoint selection, threshold calibration, and ensemble design. Blind Test set data are not used for training.

Transformer subtokens are aligned back to the original task tokens, with the boundary target attached to the final visible subtoken of each original token. Predictions are mapped back to exactly one binary label per task token. No punctuation-restoration system is used for either NoPnx subtask.

The additional NP models are trained for five epochs with learning rate 3×10^{-5}, weight decay 0.01, and batch size 8. AraELECTRA uses batch size 4 with two gradient-accumulation steps (effective batch size 8), encoder and classification-head learning rates of 2×10^{-5} and 5×10^{-5}, respectively, weight decay 0.01, 10% warm-up, and maximum gradient norm 1.0.

Our primary measure is sentence-boundary F1, reported with precision and recall. For small improvements, we additionally compare systems across document-level held-out folds and do not replace a simpler system solely for a small aggregate development gain.

Subtask	P	R	F1
PA	93.2	94.0	93.4
NoPnx-PA	87.4	89.3	87.9
NP	92.5	88.5	90.0
NoPnx-NP	83.9	87.0	84.8

Table 2: Blind Test set performance of the selected systems.

5 Results

Table 2 reports the selected Blind Test set results. According to the official AraSeg Shared Task leaderboard, Wathiq-SPU ranked 10th, 7th, 7th, and 9th in the PA, NoPnx-PA, NP, and NoPnx-NP subtasks of the Closed Track, respectively.

The results reflect the importance of explicit structural information. PA, which retains punctuation and paragraphs, achieves the highest F1, while removing punctuation has a larger effect than removing paragraph information alone; NoPnx-NP remains the hardest subtask.

For PA, the selected two-model configuration uses a default threshold of 0.543 and a lower 0.31 sentence-ending-punctuation threshold, achieving 93.2 precision, 94.0 recall, and 93.4 F1.

Independent-model ensembling improves development robustness in NoPnx-PA and NP. For NP, the selected global ensemble outperforms the earlier document-adaptive configuration in four of five held-out folds.

For NoPnx-NP, an early Blind Test set submission obtained 79.7 precision, 53.1 recall, and 62.7 F1, whereas the final CAMeLBERT–AraELECTRA system reaches 83.9 precision, 87.0 recall, and 84.8 F1, corresponding to a 33.9-point Recall increase and a 22.1-point F1 increase.

We also investigated meta-classification, excluding direct token identity and sparse character n-gram features to reduce lexical memorization and instead using contextual and structural signals, model probabilities, and model agreement. These variants did not yield sufficiently consistent held-out gains to replace the submitted neural ensembles.

Overall, calibration and probability-level ensembling remain effective across subtasks, while encoder-family diversity becomes increasingly valuable as structural cues disappear.

6 Conclusion

We presented Wathiq-SPU systems for all four subtasks of the Closed Track. Calibrated Transformer predictions, ensembling, and structural cues provide strong and reproducible segmentation. Calibration and paragraph constraints support PA, while independently trained CAMeLBERT models stabilize NoPnx-PA and NP. For NoPnx-NP, combining CAMeLBERT with AraELECTRA adds encoder-family diversity, improves recall over our early system, and yields our strongest reproducible configuration. Overall, we favor stable held-out gains over added complexity.

7 Limitations

Our systems are developed using the official AraSeg training split, with the development split used for checkpoint selection, ensemble design, and threshold calibration. Consequently, the selected weights and thresholds may not transfer directly to other datasets or substantially different genre distributions. Although AraSeg includes diverse Arabic genres, our evaluation does not establish performance on unseen corpora or dialectal settings outside the shared task.

The systems also depend on the structural information available in each subtask. Paragraph-boundary constraints can be applied when paragraph information is provided, but they cannot be used when such information is unavailable. In addition, the final ensembles require multiple Transformer inference passes, which may increase computational cost compared with a single-model system. We therefore view the reported results as evidence of effectiveness within the AraSeg evaluation setting rather than a guarantee of generalization to all Arabic text.

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